

Chewy Application Packet

Jason Dutton

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Professional Portfolio: github.com/JasonSEFIDEFI/CV- | Engineering Portfolio:

github.com/JasonSEFIDEFI/SEFI-PY | Research Portfolio: github.com/JasonSEFIDEFI/PhD

Hiring Team - Chewy

Dear Hiring Manager,

I am excited to apply for the Software Engineer II - Python position on Chewy's Supply Chain Simulation Technology team.

This opportunity immediately stood out to me because it sits at the intersection of software engineering, simulation, optimization, data transformation, and complex systems analysis. Those same themes have defined my work for years as both a Python software developer and creator of the SEFI-PY computational framework.

SEFI-PY began as an effort to build a deterministic, reproducible computational engine capable of supporting advanced simulation and modeling workflows. Through its development, I designed modular Python architectures, implemented reusable backend components, developed diagnostic and validation pipelines, and built systems focused on reliability, maintainability, and scalable experimentation.

What particularly attracts me to Chewy's Supply Chain Simulation team is the opportunity to apply these skills to real-world operational challenges at scale. Designing optimization systems, developing simulation tools, resolving data quality problems, and building engineering pipelines for deployment and testing are all areas that closely align with the work I enjoy most.

I believe my combination of Python development, computational modeling, systems engineering, technical writing, and independent research offers a unique perspective that would contribute positively to Chewy's mission.

Sincerely,
Jason Dutton

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GitHub: github.com/JasonSEFIDEFI/SEFI-PY | LinkedIn: linkedin.com/in/jason-dutton

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PROFESSIONAL SUMMARY

Software Engineer specializing in Python development, computational modeling, simulation systems, backend architecture, and reproducible engineering workflows. Creator of the SEFI-PY computational framework supporting deterministic execution, advanced modeling, diagnostic tooling, and scalable experimentation.

TECHNICAL SKILLS

Python, SQL, Bash, Backend Development, Distributed Systems, REST APIs, CI/CD, Git, Docker, Computational Modeling, Simulation Systems, Data Transformation, Optimization Frameworks, Systems Architecture.

PROFESSIONAL EXPERIENCE

Independent Theoretical Researcher & Python Software Engineer (2018-Present)

Designed and developed SEFI-PY, a Python computational framework supporting simulation, modeling, deterministic execution, and engineering experimentation.

Underground Electrical Repairman - Paramount Contura / Alpha Metallurgical Resources (2022-2024)

Performed high-reliability diagnostics and structured root-cause analysis in mission-critical environments.

Fabricator, Welder, Shop & Field Foreman - Dominion Steel (2016-2018)

Led technical teams and standardized repeatable operational processes.

Underground Electrician - Alpha Natural Resources (2010-2016)

Maintained complex electrical infrastructure and executed reliability-focused troubleshooting.

PROJECTS

SEFI-PY Computational Framework

Modular Python engine emphasizing deterministic execution, reproducibility, simulation workflows, and maintainable architecture.

Computational Modeling & Simulation Research

Built reproducible simulation-oriented workflows and validation pipelines for complex-system analysis.

RESEARCH PORTFOLIO

Geometric Worldline Foundations of the Electron Field; SEFI: The Single Entity Field Interpretation; Unified Geometric Field Theory from Worldline and Identity-Space Invariants; Geometric Photonic Quantum Error Correction via SEFI, GWFM, and DEFI; Dynamic Entity Field Integration (DEFI).

EDUCATION

Independent Research Portfolio in Computational Physics, Applied Mathematics, Geometric Field Theory, Simulation Framework Design, and Quantum Error Correction.

ACADEMIC DEVELOPMENT

Pursuing academic research development associated with the University of Westminster, supporting publication-oriented work in computational physics, mathematical modeling, and geometric field theory. Research manuscripts are being prepared and submitted for peer-reviewed publication.